Dynamics of a rigid body in space

' ... beato te, Giorgio, che ai tuoi studenti parli dei sistemi multi-body! Ti confesso che i miei, quando gli faccio la dinamica nello spazio fatta bene, hanno dei problemi già con il single-body'
Prof. Furio Vatta al Prof. Giorgio Diana, III Seminario di Meccanica Applicata alle Macchine, Lucca, 1995

1 Introduction

This lesson aims at providing some basic concepts concerning the spatial kinematics and dynamics of a rigid body. Important new concepts are introduced and differences outlined in respect to the case of planar motion for a rigid body, particularly the body's inertia tensor and non-commutativity of rotations.

2 Kinetic energy of a rigid body in spatial motion.

We start by recalling the Rivals theorem stating that, given any two points *P* and *Q* on a same rigid body, the velocity vector $\vec{v}_P$ for point *P* can be expressed as:

$$\vec{v}_P = \vec{v}_Q + \vec{\omega} \wedge (P - Q) \tag{2.1}$$

with $\vec{v}_Q$ the velocity vector for point *Q*, $\vec{\omega}$ the angular speed of the body and $(P-Q)$ the vector defining the position of point *P* with respect to point *Q*. This theorem was widely used in exercises across the whole of this course considering the planar motion of rigid bodies, but its validity is not restricted to the planar motion case. If we use the body centre of mass (c.o.m.) *G* instead of the generic pint *P*, Eq. (2.1) becomes:

$$\vec{v}_P = \vec{v}_G + \vec{\omega} \wedge (P - G) \tag{2.2}$$

We now introduce a formalism to rewrite Eq.s (2. 2) in matrix form. To this aim, we introduce a Cartesian reference centred in the body c.o.m. (so that point *G* is the origin of the reference) and we define the following column vectors:

$$\underline{v}_P = \begin{Bmatrix} v_{Px} \\ v_{Py} \\ v_{Pz} \end{Bmatrix} \; ; \; \underline{v}_G = \begin{Bmatrix} v_{Gx} \\ v_{Gy} \\ v_{Gz} \end{Bmatrix} \; ; \; \underline{\omega} = \begin{Bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{Bmatrix} \; ; \; \underline{x}_{PG} = \begin{Bmatrix} x_P \\ y_P \\ z_P \end{Bmatrix} \tag{2.3}$$

containing the components along the axis of the reference of all geometric vectors appearing in Eq. (2.2.). It is easy to show that the Cartesian components of vector $\underline{v}_P$ resulting from Eq. (2.2.) are defined by the following matrix expression, representing the matrix form of the Rival's theorem:

$$\underline{v}_P = \underline{v}_G + [\tilde{x}_{PG}]\underline{\omega} \tag{2.4}$$

with $[\tilde{x}_{PG}]$ a skew-symmetric matrix defined as follows.

$$[\tilde{x}_{PG}] = \begin{bmatrix} 0 & z_P & -y_P \\ -z_P & 0 & x_P \\ y_P & -x_P & 0 \end{bmatrix} \tag{2.5}$$

being *x*, *y* and *z* the Cartesian components of the distance of point *P* from point *G* in the chosen reference frame.
We now use Eq.s (2.4-5) to obtain an expression for the kinetic energy of a rigid body in spatial motion: for any generic body (not limited to the rigid body case), the kinetic energy reads:

$$T = \frac{1}{2} \int_{\mathrm{Vol}} \underline{v}_P^T \underline{v}_P \rho dVol \tag{2.6}$$

Replacing (2.4) and (2.5) in (2.6) we obtain an expression of the Kinetic energy valid only for a rigid body:

$$T = \frac{1}{2} \underline{v}_G^T \underbrace{\int_{\mathrm{Vol}} \rho dVol}_{m} \underline{v}_G + \frac{1}{2} \underline{v}_G^T \underbrace{\int_{\mathrm{Vol}} \rho [\tilde{x}_{PG}] dVol}_{[0]} \underline{\omega} + \frac{1}{2} \underline{\omega}^T \underbrace{\int_{\mathrm{Vol}} \rho [\tilde{x}_{PG}]^T dVol}_{[0]} \underline{v}_G + \frac{1}{2} \underline{\omega}^T \underbrace{\int_{\mathrm{Vol}} \rho [\tilde{x}_{PG}]^T [\tilde{x}_{PG}] dVol}_{[J_G]} \omega \tag{2.7}$$

Regarding Eq. (2.7), we note the following:

- in the first term, the integral equals the total mass *m* of the body;
- the second and third terms a square matrix of order n=3 whose elements are the first-order mass moments of the body with respect to point G, i.e. the body's c.o.m. By definition, these terms are therefore zero;
- the integral appearing in the fourth term is also a square matrix of order n=3 $[J_G]$. On account of (2.5) this matrix reads[1]:

$$[J_G] = \int_{\mathrm{Vol}} \rho [\tilde{x}_{PG}]^T [\tilde{x}_{PG}] dVol = \int_{\mathrm{Vol}} \rho \begin{bmatrix} y^2 + z^2 & -xy & -xz \\ -xy & z^2 + x^2 & -yz \\ -xz & -yz & x^2 + y^2 \end{bmatrix} dVol \tag{2.8}$$

The matrix defined in (2.8) is a symmetric matrix. The diagonal terms are easily recognised as the mass moments of inertia of the body with respect to the *x*, *y*, *z* axes respectively.

$$J_{xx} = \int_{\mathrm{Vol}} \rho \left(y^2 + z^2\right) dVol \;\; ; \;\; J_{yy} = \int_{\mathrm{Vol}} \rho \left(z^2 + x^2\right) dVol \;\; ; \;\; J_{zz} = \int_{\mathrm{Vol}} \rho \left(x^2 + y^2\right) dVol \tag{2.9}$$

The off-diagonal terms are called *products of inertia*, defined as:

$$J_{xy} = J_{yx} = -\int_{\mathrm{Vol}} \rho (xy) dVol \;\; ; \;\; J_{yz} = J_{zy} = -\int_{\mathrm{Vol}} \rho (yz) dVol \;\; ; \;\; J_{zx} = J_{xz} = -\int_{\mathrm{Vol}} \rho (zx) dVol \tag{2.10}$$

The terms defined by Eq.s (2.9-10) fully define the inertial properties of the body in the chosen reference frame. The whole matrix is called the body's *inertia tensor*. According to the notation, introduced in (2.9-10), this matrix reads:

$$[J_G] = \begin{bmatrix} J_{xx} & J_{xy} & J_{xz} \\ J_{yx} & J_{yy} & J_{yz} \\ J_{zx} & J_{zy} & J_{zz} \end{bmatrix} \tag{2.11}$$

Quite obviously, the value of the moments and products of inertia are changing with the choice of the *x-y-z* reference. Regardless the shape and mass distribution of the rigid body, it is always possible to identify one 'privileged' Cartesian reference in which the products of inertia reduce to zero and $[J_G]$ is a diagonal matrix. The axis of this reference are called the *principal axis of inertia* of the body. If the body has three orthogonal planes of symmetry, the principal axis of inertia of the body are defined by the intersections of the symmetry planes.

The expression of the kinetic energy for the single body can be written in its final form as:

[1] For the sake of simplicity, we drop subscript P to the coordinates x, y, z in the expressions below.

$$T = \frac{1}{2} m \underline{v}_G^T \underline{v}_G + \frac{1}{2} \underline{\omega}^T \left[J_G\right] \omega \qquad (2.12)$$

Eq. (2.12) is known as the König's theorem and represents a generalisation to 3D of the known expression of the kinetic energy for a rigid body moving in a plane.

3 Global and local reference frames.

The expressions obtained in Section 2 for the kinetic energy and for the parameters describing the inertia of the body are valid regardless the choice of the reference frame. In our study, we will mostly use two reference frames:

i) a *global* reference, i.e. a reference that is not moving, so that it can be used to describe the absolute motion of the body
ii) a *local* reference, moving together with the body, so that the speed of any point in the body will be zero for an observer located in this reference.

The origin of the local reference is often chosen to coincide with the body c.o.m., but there are cases when it is more advantageous to have the origin of this reference not coinciding with the c.o.m., see e.g. the example in Section 6 of this note.
From now on, we will use superscript "G" to denote the Cartesian components of a vector in the global reference (i.e. $\underline{v}_G^G$ for the components of the body's c.o.m. in the global reference), and a superscript "L" to denote the components of a vector in the local reference (i.e. $\underline{v}_G^L$ for the components of the body's c.o.m. in the global reference).
Equation (2.12) can be written using the components of the angular speed and body's c.o.m. velocity in any of these two references (or even according to a different reference). It is clear however that the volume integrals defining the moments and products of inertia will be independent from time only when these are evaluated in the local reference.

4 Angular position and angular speed of a rigid body moving in a 3D space.

The kinematics of a rigid body in spatial movement is significantly more complex than in the 2D case, mostly on account of rotations not being commutative. This means in particular that suitable coordinates need to be introduced to describe the orientation of the body in the 3D space. Different possible choices are commonly in use, see [1] for a comprehensive treatment. For the sake of brevity we will confine these notes to one single set of coordinates, known as the *Cardan angles*.
The Cardan's angles consist of three consecutive rotations that applied in a prescribed sequence transform the directions of the global reference axes into the directions of the local reference axes. The three rotations are (cf. Figure 4.1):

1. a rotation σ performed around the x^G axis of the global reference that transforms the global reference x^G-y^G-z^G in the axes of an intermediate reference "I" x^I-y^I-z^I.
2. a rotation β performed around the y^I axis that transforms the axes of the intermediate reference "I" x^I-y^I-z^I in the axes of a second intermediate reference II x^{II}-y^{II}-z^{II}.
3. a rotation ρ performed around the z^{II} axis that transforms the axes of the intermediate reference II x^{II}-y^{II}-z^{II} in the axes of the local reference x^L-y^L-z^L.

The three angles σ, β, ρ are called the Cardan angles and their value univocally defines the orientation of the local reference with respect to the global reference. We note that, because rotations in space are non-commutative, the definition of the Cardan angles is associated with the order of rotations 1. -2. -3. defined above: applying the rotation in a different order would result in a different orientation of the local reference.

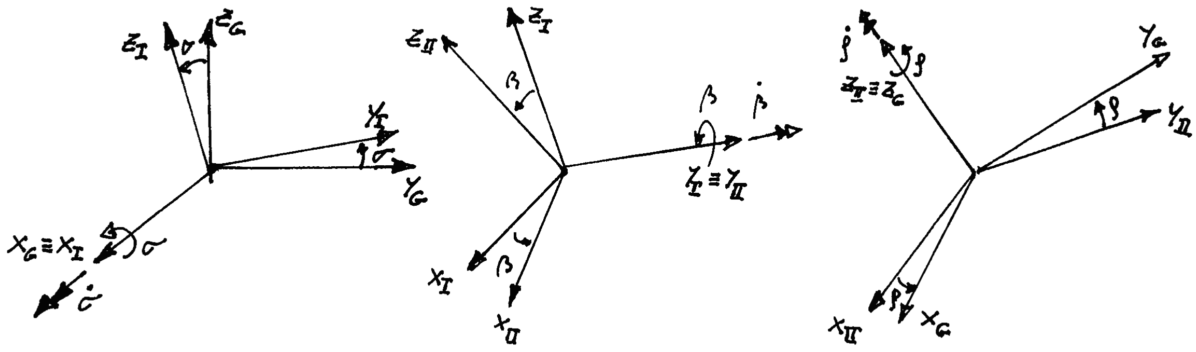

Figure 4-1 The Cardan angles

4.1 Expression of the rotation matrix as function of the Cardan angles.

The rotation matrix $[\Lambda_{LG}]$ is a matrix defining the transformation of the components of a vector along the global reference $\underline{a}^G$ in the components of the same vector along the local reference $\underline{a}^L$:

$$\underline{a}^L = [\Lambda_{LG}]\underline{a}^G \tag{4.1}$$

This matrix depends on the orientation of the local reference with respect to the global one and therefore, ultimately, from the value of the Cardan angles. In order to define the expression of the rotation matrix as function of the Cardan angles, we proceed according to the three successive rotations defined above. First, we define a rotation matrix $[\Lambda_{I-G}]$ transforming the vector components from the global reference to reference intermediate I:

$$\underline{a}^I = [\Lambda_{I-G}]\underline{a}^G \tag{4.2}$$

this is the well-known planar rotation matrix corresponding to a rotation performed around the x_G axis, and is a function of the Cardan angle σ:

$$[\Lambda_{I-G}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\sigma & \sin\sigma \\ 0 & -\sin\sigma & \cos\sigma \end{bmatrix} \tag{4.3}$$

In a similar way it is possible to define the planar rotation matrices $[\Lambda_{II-I}]$ and $[\Lambda_{L-II}]$, transforming the components of a vector respectively from the intermediate I to the intermediate II references and from the intermediate II to the local reference:

$$\underline{a}^{II} = [\Lambda_{II-I}]\underline{a}^I \quad ; \quad \underline{a}^L = [\Lambda_{L-II}]\underline{a}^{II} \tag{4.4}$$

these matrices are function of the Cardan angles β and ρ respectively:

$$[\Lambda_{II-I}] = \begin{bmatrix} \cos\beta & 0 & -\sin\beta \\ 0 & 1 & 0 \\ \sin\beta & 0 & \cos\beta \end{bmatrix} \quad ; \quad [\Lambda_{L-II}] = \begin{bmatrix} \cos\rho & \sin\rho & 0 \\ -\sin\rho & \cos\rho & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{4.5}$$

Using (4.2) and (4.4) we get:

$$\underline{a}^L = [\Lambda_{L-II}]\underline{a}^{II} = [\Lambda_{L-II}][\Lambda_{II-I}]\underline{a}^I = [\Lambda_{L-II}][\Lambda_{II-I}][\Lambda_{I-G}]\underline{a}^G \tag{4.6}$$

so we obtain:

$$[\Lambda_{LG}] = [\Lambda_{L-II}][\Lambda_{II-I}][\Lambda_{I-G}] \tag{4.7}$$

We note that the planar rotation matrices defined by (4.3) and (4.5) are orthogonal, so that their inverse matrix coincides with the transpose (note that transposing the planar rotation matrix corresponds to changing the sign of the angle defining the planar rotation). Because a matrix obtained as a product of orthogonal matrices is orthogonal, it follows that matrix $[\Lambda_{LG}]$ is also orthogonal:

$$[\Lambda_{LG}]^{-1} = [\Lambda_{LG}]^T \tag{4.8}$$

We also note that the relationship transforming the components of a vector from the local to the global reference is:

$$\underline{a}^G = [\Lambda_{GL}]\underline{a}^L \tag{4.9}$$

with:

$$[\Lambda_{GL}] = [\Lambda_{LG}]^{-1} = [\Lambda_{LG}]^T \tag{4.10}$$

4.2 Expression of the angular speed of the rigid body as function of the Cardan's angles and their time derivatives.

The angular speed vector is defined as the ratio of the infinitesimal rotation produced by an infinitesimal increase dσ, dβ, dρ of the Cardan angles, over the infinitesimal time dt over which the infinitesimal rotation takes place. Owing to the definition of the Cardan angles, it follows that the angular speed vector is the sum of the following three terms:

1. a vector with intensity $\dot{\sigma}$ pointing along the x^G axis of the global reference, coinciding with the axis x^I of the intermediate reference I;
2. a vector with intensity $\dot{\beta}$ pointing along the y^I axis of the intermediate reference I, coinciding with the y^{II} axis of the intermediate reference II;
3. a vector with intensity $\dot{\rho}$ pointing along the z^{II} axis of the intermediate reference II, coinciding with the z^L axis of the local reference.

We are interested in defining the components of the angular speed vector along the axes of the global and local reference. First, we consider the projections of the vector in the global reference:

$$\underline{\omega}^G = \begin{Bmatrix} \dot{\sigma} \\ 0 \\ 0 \end{Bmatrix} + [\Lambda_{G-I}]\left(\begin{Bmatrix} 0 \\ \dot{\beta} \\ 0 \end{Bmatrix} + [\Lambda_{I-II}]\begin{Bmatrix} 0 \\ 0 \\ \dot{\rho} \end{Bmatrix}\right) \tag{4.11}$$

Replacing in (4.11) the expressions for matrices $[\Lambda_{G-I}]$ and $[\Lambda_{I-II}]$ and re-arringing the result, we obtain:

$$\underline{\omega}^G = \begin{bmatrix} 1 & 0 & \sin\beta \\ 0 & \cos\sigma & -\sin\sigma\cos\beta \\ 0 & \sin\sigma & \cos\sigma\cos\beta \end{bmatrix}\begin{Bmatrix} \dot{\sigma} \\ \dot{\beta} \\ \dot{\rho} \end{Bmatrix} = [A^G]\underline{\dot{\theta}} \tag{4.12}$$

with $\underline{\dot{\theta}}$ the vector collecting the time derivatives of the three Cardan angles and $[A^G]$ a matrix transforming the time derivatives of the Cardan angles in the components of the angular speed vector along the global reference.

A similar procedure can be followed to define the components of the angular speed vector along the local reference $\underline{\omega}^L$:

$$\underline{\omega}^L = \begin{Bmatrix} 0 \\ 0 \\ \dot{\rho} \end{Bmatrix} + \left[\Lambda_{L-II}\right] \left(\begin{Bmatrix} 0 \\ \dot{\beta} \\ 0 \end{Bmatrix} + \left[\Lambda_{II-I}\right] \begin{Bmatrix} \dot{\sigma} \\ 0 \\ 0 \end{Bmatrix} \right) \tag{4.13}$$

which results in:

$$\underline{\omega}^L = \begin{bmatrix} \cos\rho\cos\beta & \sin\rho & 0 \\ -\sin\rho\cos\beta & \cos\rho & 0 \\ \sin\beta & 0 & 1 \end{bmatrix} \begin{Bmatrix} \dot{\sigma} \\ \dot{\beta} \\ \dot{\rho} \end{Bmatrix} = \left[A^L\right]\dot{\underline{\theta}} \tag{4.12}$$

5 Dynamics of a rigid body in space – some fundamentals

The equations describing the motion of a rigid body in space can be conveniently written using the Lagrange equations. An alternative approach would be to use Newtonian dynamics, or equivalently the D'Alembert's principle, but this would require extending to the 3D case the equations defined in Lesson 1 for the planar motion of the body, which is not trivial and cannot be dealt with here due to lack of time.
In this section, an outline of the derivation of the equations of motion for a single rigid body is presented, using the Lagrange equations. For the sake of simplicity, the case of an unconstrained rigid body is considered, whereas the presence of constraints is dealt with in a simple case shown as an example in Section 6.

5.1 Independent coordinates of an unconstrained rigid body moving in a 3D space.

In order to define the position of a rigid body in a 3D space, it is necessary to define the position of one point in the body and the orientation of the body. If the body is not subject to constraints, it follows that a total of six independent coordinates are needed to define the position of the bodies: the three coordinates of a point O (e.g. the origin of the local reference) and three independent kinematic parameters (e.g. the Cardan angles) defining the orientation of the body. The independent coordinates of the body are collected in vector $\underline{x}$, whereas the coordinates of point O, also called *translational coordinates* of the body, are collected in vector $\underline{x}_O^G$ while the Cardan angles, also called *rotational coordinates* of the body, are collected in vector $\underline{\theta}$:

$$\underline{x} = \begin{Bmatrix} x_O^G \\ y_O^G \\ z_O^G \\ \sigma \\ \beta \\ \rho \end{Bmatrix} = \begin{Bmatrix} \underline{x}_O^G \\ \underline{\theta} \end{Bmatrix} \;;\; \underline{x}_O = \begin{Bmatrix} x_O^G \\ y_O^G \\ z_O^G \end{Bmatrix} \;;\; \underline{\theta} = \begin{Bmatrix} \sigma \\ \beta \\ \rho \end{Bmatrix} \tag{5.1}$$

5.2 Expression of the body's kinetic energy as a function of the independent coordinates.

Based on the results obtained in Sections 2 and 4, it is possible to write the expression of the kinetic energy of the body as a function of the independent coordinates, which is then derived according to Lagrange equations to obtain the inertial terms in the equations of motion of the body.
We start by rewriting (2.12) as:

$$T = \tfrac{1}{2}\underline{v}_G^{G^T}[m]\underline{v}_G^G + \tfrac{1}{2}\underline{\omega}^{L^T}\left[J_G^L\right]\underline{\omega}^L \tag{5.2}$$

In this expression, the components of the c.o.m. velocity along the global reference are used to define the translational component of the kinetic energy, whereas the components of the angular speed along the local reference are used to define the rotational component of the kinetic energy: in this way, the body's tensor of inertia defined in the local reference $\left[J_G^L\right]$ is used, whose elements are constant with time and only depend on the distribution of masses in the body. Matrix $[m]$ appearing in (5.2) is a diagonal matrix with the mass of the body in all diagonal positions.
Equation (5.2) can be rewritten in the form of a single matrix product as follows:

$$T = \tfrac{1}{2}\begin{Bmatrix}\underline{v}_G^G \\ \underline{\omega}^L\end{Bmatrix}^T \begin{bmatrix}[m] & [0] \\ [0] & \left[J_G^L\right]\end{bmatrix}\begin{Bmatrix}\underline{v}_G^G \\ \underline{\omega}^L\end{Bmatrix} \tag{5.3}$$

Assuming point O not being the body's c.o.m., the following expression is obtained for the components of the c.o.m. speed in the global reference:

$$\underline{v}_G^G = \underline{v}_O^G + \left[\Lambda_{GL}\right]\left[\tilde{x}_G^L\right]\underline{\omega}^L = \underline{\dot{x}}_o^G + \left[\Lambda_{GL}\right]\left[\tilde{x}_G^L\right]\left[A^L\right]\underline{\dot{\theta}} \tag{5.4}$$

Therefore, the following relationship is obtained relating the kinematic quantities appearing in (5.2) and the time derivatives of the body's independent coordinates:

$$\begin{Bmatrix}\underline{v}_G^G \\ \underline{\omega}^L\end{Bmatrix} = \begin{bmatrix}[I] & \left[\Lambda_{GL}\right]\left[\tilde{x}_G^L\right]\left[A^L\right] \\ [0] & \left[A^L\right]\end{bmatrix}\begin{Bmatrix}\underline{\dot{x}}_o^G \\ \underline{\dot{\theta}}\end{Bmatrix} \tag{5.5}$$

Substituting (5.5.) in (5.3) the following final expression of the kinetic energy is obtained:

$$T = \tfrac{1}{2}\begin{Bmatrix}\underline{\dot{x}}_o^G \\ \underline{\dot{\theta}}\end{Bmatrix}^T \begin{bmatrix}[m] & [m]\left[\Lambda_{GL}\right]\left[\tilde{x}_G^L\right]\left[A^L\right] \\ \left[A^L\right]^T\left[\tilde{x}_G^L\right]^T\left[\Lambda_{GL}\right]^T[m] & \left[A^L\right]^T\left(\left[\tilde{x}_G^L\right]^T[m]\left[\tilde{x}_G^L\right] + \left[J_G^L\right]\right)\left[A^L\right]\end{bmatrix}\begin{Bmatrix}\underline{\dot{x}}_o^G \\ \underline{\dot{\theta}}\end{Bmatrix} = \tfrac{1}{2}\underline{\dot{x}}^T[M]\underline{\dot{x}} \tag{5.6}$$

with $[M]$ the 6x6 mass matrix of the body.
In (5.6) the term appearing in the 3x3 lower right block of the mass matrix can be interpreted as the tensor of inertia of the body evaluated with respect to centre O:

$$\left[J_O^L\right] = \left[\tilde{x}_G^L\right]^T[m]\left[\tilde{x}_G^L\right] + \left[J_G^L\right] \tag{5.7}$$

Considering the special case of point O being the body's c.o.m. ($O \equiv G$), the following simplified expression of the body's kinetic energy and mass matrix are obtained:

$$T = \tfrac{1}{2}\begin{Bmatrix}\underline{\dot{x}}_G^G \\ \underline{\dot{\theta}}\end{Bmatrix}^T \begin{bmatrix}[m] & [0] \\ [0] & \left[A^L\right]^T\left[J_G^L\right]\left[A^L\right]\end{bmatrix}\begin{Bmatrix}\underline{\dot{x}}_G^G \\ \underline{\dot{\theta}}\end{Bmatrix} \tag{5.8}$$

6 An example: the dynamics of a rigid body rotating around a fixed axis.

We provide here an example in which the equations developed throughout Sections 2-5 are used. The case considered consists of a single rigid body rotating around a fixed axis. The system is shown in Figure 6-1: a rigid body with generic mass distribution is subjected to constraints consisting of a spherical hinge located at point A and a double cart at point B. A spherical hinge is a constraint preventing all displacement components at the point where it is applied, whereas the double cart prevents 2 out of the 3 components of displacement at the point where it is applied.

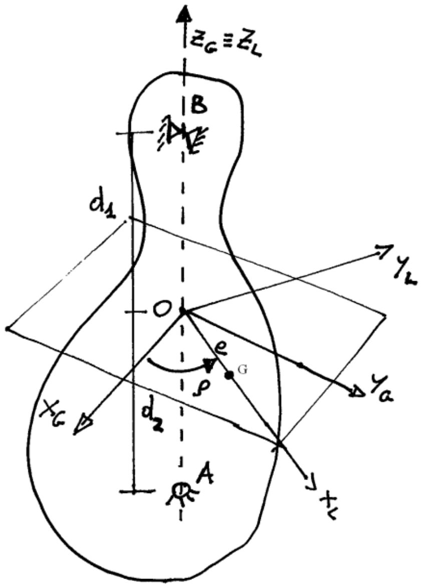

Figure 6-1 Rigid body rotating around a fixed axis

Because of the constraints applied, the rigid body has 1 degree of freedom, consisting of the rotation around a fixed axis passing through points A and B. This is a known kind of motion and all points in the body perform a planar motion, corresponding to a trajectory which is a circumference laying in a plane perpendicular to the axis of rotation. However, due to the generic mass distribution of the body, the inertia forces generated by the rotation are not balanced so that ultimately the significant issue with this example is to compute the constraint forces at A and B as a function of the parameters defining the inertial properties of the body, rather than in the study of the motion itself.
The global and local references are chosen as follows: the origin O of both references are located at the intersection of the axis of rotation with a plane perpendicular to the axis of rotation and passing through the body c.o.m. G. The x_G and y_G axes of the global reference are taken as two arbitrary fixed and orthogonal directions in the plane defined above and the z_G axis coincides with the axis of rotation. The x_L axis is chosen as the axis pointing from O to G, y_L is perpendicular to x_L and lies in the same plane as x_G and y_G. Finally, the z_L axis coincides with z_G. Owing to the special case considered, the Cardan angles σ and β have zero value at any time, and the position of the body is defined by the single independent coordinate ρ. We nevertheless consider as the coordinates of the body the three translational and three rotational coordinates defined by (5.1) and we perform a partition of vector $\underline{x}$ into vectors $\underline{x}_c$ and $\underline{x}_f$, representing respectively the coordinates subjected to constraints and the single free coordinate:

$$\underline{x} = \begin{Bmatrix} x_O^G \\ y_O^G \\ z_O^G \\ \sigma \\ \beta \\ \rho \end{Bmatrix} = \begin{Bmatrix} \underline{x}_c \\ \underline{x}_f \end{Bmatrix} \ ; \ \underline{x}_c = \begin{Bmatrix} x_O^G \\ y_O^G \\ z_O^G \\ \sigma \\ \beta \end{Bmatrix} = \underline{0} \ ; \ \underline{x}_f = \{\rho\} \tag{6.1}$$

We also note that point O is still, so the three translational coordinates of the system are zero at any time, so that the whole vector $\underline{x}_c$ is zero at any time.

By writing Lagrange equations in matrix form considering the entire vector $\underline{x}$ we obtain not only the single equation of motion for the body, but also five equations that can be used to compute the constraint forces at points A and B.

Assuming that no external force and no gravitational force acts on the body, Lagrange equations in matrix form read:

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \underline{\dot{x}}}\right)^T - \left(\frac{\partial T}{\partial \underline{x}}\right)^T = \underline{Q}_c \tag{6.2}$$

with $\underline{Q}_c$ the vector of the Lagrangian components of the constraint forces.

6.1 Lagrangian components of the constraint forces.

First of all, we look for an expression of vector $\underline{Q}_c$. To this aim, we first write the virtual work of the reaction forces :

$$\delta W_c = \delta \underline{x}_A^{G^T} \underline{R}_A^G + \delta \underline{x}_B^{G^T} \underline{R}_B^G \tag{6.3}$$

with:

$$\underline{R}_A^G = \begin{Bmatrix} R_{Ax}^G \\ R_{Ay}^G \\ R_{Az}^G \end{Bmatrix} \; ; \; \underline{R}_B^G = \begin{Bmatrix} R_{Bx}^G \\ R_{By}^G \\ 0 \end{Bmatrix} \tag{6.4}$$

the vectors defining the components along the global reference of the constraint forces applied at points A and B respectively. Note that the double cart at point B only applies constraint forces along the x_G and y_G directions.

The virtual displacements of the body evaluated at points A and B are defined according to:

$$\delta \underline{x}_A^G = \delta \underline{x}_O^G + \left[\tilde{x}_A^G\right]\left[A^G\right]\delta\underline{\theta} \; ; \; \delta \underline{x}_B^G = \delta \underline{x}_O^G + \left[\tilde{x}_B^G\right]\left[A^G\right]\delta\underline{\theta} \tag{6.5}$$

Replacing (6.5) in (6.3) and reworking properly, the following is obtained:

$$\delta W_c = \begin{Bmatrix} \delta \underline{x}_O^G \\ \delta\underline{\theta} \end{Bmatrix}^T \begin{Bmatrix} \underline{R}_A^G + \underline{R}_B^G \\ \left[A^G\right]^T \left(\left[\tilde{x}_A^G\right]^T \underline{R}_A^G + \left[\tilde{x}_B^G\right]^T \underline{R}_B^G\right) \end{Bmatrix} = \delta \underline{x}^T \underline{Q}_c \tag{6.6}$$

Considering σ=0 and β=0, we note that matrix $\left[A^G\right]$ defined by (4.12) becomes the identity matrix. Furthermore, considering the value of the coordinates of points A and B in the global reference, we get:

$$\left[\tilde{x}_A^G\right] = \begin{bmatrix} 0 & -d_1 & 0 \\ d_1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \; ; \; \left[\tilde{x}_B^G\right] = \begin{bmatrix} 0 & d_2 & 0 \\ -d_2 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{6.7}$$

replacing (6.7) in (6.6) and recalling that $\left[A^G\right] = [I]$ we get the final expression for vector $\underline{Q}_c$:

$$\underline{Q}_c = \begin{Bmatrix} R_{Ax}^G + R_{Bx}^G \\ R_{Ay}^G + R_{By}^G \\ R_{Az}^G \\ d_1 R_{Ay}^G - d_2 R_{By}^G \\ -d_1 R_{Ax}^G + d_2 R_{Bx}^G \\ 0 \end{Bmatrix} \tag{6.8}$$

We note that the first three elements in vector $\underline{Q}_c$ are the components of the resultant of the constraint forces, whereas the other three terms are the moments of the constraint forces with respect to the three axes of the global reference. Because all the constraint force components are passing through the z_G axis, the last term in vector $\underline{Q}_c$ is zero.

6.2 *Equation of motion and reaction forces.*

We now apply Lagrange equations according to (6.2).

$$\left(\frac{\partial T}{\partial \underline{\dot{x}}}\right)^T = \begin{bmatrix} [m] & [m][\Lambda_{GL}]\left[\tilde{x}_G^L\right]\left[A^L\right] \\ \left[A^L\right]^T\left[\tilde{x}_G^L\right]^T[\Lambda_{GL}]^T[m] & \left[A^L\right]^T\left(\left[\tilde{x}_G^L\right]^T[m]\left[\tilde{x}_G^L\right]+\left[J_G^L\right]\right)\left[A^L\right] \end{bmatrix} \begin{Bmatrix} \underline{\dot{x}}_o^G \\ \underline{\dot{\theta}} \end{Bmatrix} \tag{6.9}$$

Because the body c.o.m. is on the x_L axis at a distance e from the origin O, we have:

$$\left[\tilde{x}_G^L\right] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & e \\ 0 & -e & 0 \end{bmatrix} \tag{6.10}$$

and therefore:

$$\left[J_O^L\right] = \left[\tilde{x}_G^L\right]^T[m]\left[\tilde{x}_G^L\right] + \left[J_G^L\right] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & me^2 & 0 \\ 0 & 0 & me^2 \end{bmatrix} + \begin{bmatrix} J_{Gxx}^L & J_{Gxy}^L & J_{Gxz}^L \\ J_{Gyx}^L & J_{Gyy}^L & J_{Gyz}^L \\ J_{Gzx}^L & J_{Gzy}^L & J_{Gzz}^L \end{bmatrix} \tag{6.11}$$

this is a constant matrix, only depending on the mass distribution of the body.
Furthermore, we define matrix $[E]$ as:

$$[E] = [m][\Lambda_{GL}]\left[\tilde{x}_G^L\right]\left[A^L\right] \tag{6.12}$$

According to (6.11) and (6.12) we rewrite (6.9) in the form:

$$\left(\frac{\partial T}{\partial \underline{\dot{x}}}\right)^T = \begin{Bmatrix} [m]\underline{\dot{x}}_O^G + [E]\underline{\dot{\theta}} \\ [E]^T \underline{\dot{x}}_O^G + \left[A^L\right]^T\left[J_O^L\right]\left[A^L\right]\underline{\dot{\theta}} \end{Bmatrix} \tag{6.13}$$

Considering (6.1) we have:

$$\underline{\dot{x}}_O^G = \underline{0} \;\; ; \;\; \underline{\dot{\theta}} \begin{Bmatrix} 0 \\ 0 \\ \dot{\rho} \end{Bmatrix} \tag{6.14}$$

and:

$$[\Lambda_{GL}] = \begin{bmatrix} \cos\rho & -\sin\rho & 0 \\ \sin\rho & \cos\rho & 0 \\ 0 & 0 & 1 \end{bmatrix} \; ; \; [A^L] = \begin{bmatrix} \cos\rho & \sin\rho & 0 \\ -\sin\rho & \cos\rho & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{6.15}$$

Considering (6.10) and (6.15) the expression (6.12) for matrix $[E]$ becomes:

$$[E] = me \begin{bmatrix} 0 & 0 & -\sin\rho \\ 0 & 0 & \cos\rho \\ \sin\rho & -\cos\rho & 0 \end{bmatrix} \tag{6.16}$$

Furthermore, because of (6.14) and (6.15) we have:

$$[A^L]\dot{\underline{\theta}} = \dot{\underline{\theta}} \tag{6.17}$$

Considering (6.14-17), (6.13) becomes:

$$\left(\frac{\partial T}{\partial \dot{\underline{x}}}\right)^T = \begin{Bmatrix} [E]\dot{\underline{\theta}} \\ [A^L]^T [J_O^L]\dot{\underline{\theta}} \end{Bmatrix} = \begin{Bmatrix} -me\sin\rho \\ me\cos\rho \\ 0 \\ J_{Oxz}^L\cos\rho - J_{Oyz}^L\sin\rho \\ J_{Oxz}^L\sin\rho + J_{Oyz}^L\cos\rho \\ J_{Ozz}^L \end{Bmatrix}\dot{\rho} \tag{6.13}$$

and:

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{\underline{x}}}\right)^T = \begin{Bmatrix} -me\sin\rho \\ me\cos\rho \\ 0 \\ J_{Oxz}^L\cos\rho - J_{Oyz}^L\sin\rho \\ J_{Oxz}^L\sin\rho + J_{Oyz}^L\cos\rho \\ J_{Ozz}^L \end{Bmatrix}\ddot{\rho} + \begin{Bmatrix} -me\cos\rho \\ -me\sin\rho \\ 0 \\ -J_{Oxz}^L\sin\rho - J_{Oyz}^L\cos\rho \\ J_{Oxz}^L\cos\rho - J_{Oyz}^L\sin\rho \\ 0 \end{Bmatrix}\dot{\rho}^2 \tag{6.14}$$

Finally, we note that, given that all elements in vector $\dot{\underline{x}}$ are zero except the last one, we have:

$$-\left(\frac{\partial T}{\partial \underline{x}}\right)^T = -\left(\frac{\partial \frac{1}{2} J_{Ozz}^L \dot{\rho}^2}{\partial \underline{x}}\right)^T = \underline{0} \tag{6.15}$$

Therefore, the following set of equations is obtained from (6.2):

$$\begin{Bmatrix} -me\sin\rho \\ me\cos\rho \\ 0 \\ J_{Oxz}^L\cos\rho - J_{Oyz}^L\sin\rho \\ J_{Oxz}^L\sin\rho + J_{Oyz}^L\cos\rho \\ J_{Ozz}^L \end{Bmatrix}\ddot{\rho} + \begin{Bmatrix} -me\cos\rho \\ -me\sin\rho \\ 0 \\ -J_{Oxz}^L\sin\rho - J_{Oyz}^L\cos\rho \\ J_{Oxz}^L\cos\rho - J_{Oyz}^L\sin\rho \\ 0 \end{Bmatrix}\dot{\rho}^2 = \begin{Bmatrix} R_{Ax}^G + R_{Bx}^G \\ R_{Ay}^G + R_{By}^G \\ R_{Az}^G \\ d_1 R_{Ay}^G - d_2 R_{By}^G \\ -d_1 R_{Ax}^G + d_2 R_{Bx}^G \\ 0 \end{Bmatrix} \tag{6.16}$$

6.3 Physical interpretation of the equations of motion.

Equations (6.16) were obtained using Lagrange equations. However, they can be given a physical interpretation using Newton's 2nd law or the D'Alembert's principle. In more detail, the first three equations can be seen as resulting from the dynamic equilibrium of the body to translations along the x_G, y_G and z_G directions respectively: the vector sum of the inertia forces gives rise to a tangential inertia force (proportional to $\ddot{\rho}$), and to a centrifugal force (proportional to $\dot{\rho}^2$), both acting in a plane perpendicular to the body's axis of rotation. We note that when the eccentricity e of the body's centre of mass reduces to zero both these inertia forces disappear. In other words, when the body's c.o.m. is located on the axis of rotation the resultant of inertia forces for the entire body is zero. We also note that under the assumptions of this study there is no force acting along the z_G direction apart from the constraint force at the spherical hinge (point A).

Considering now the last three scalar equations resulting from (6.16), we recognise these can be given a physical interpretation as the dynamic equilibrium of the moments acting respectively along axes x_G, y_G and z_G. The last one reduces to:

$$J_{Ozz}^{L}\ddot{\rho} = 0 \tag{6.17}$$

This is the equation describing the rotation $\rho=\rho(t)$ of the body around the z_G axis. Because no external forces are included in the analysis, this equation shows that the body performs a rotation at constant angular speed $\dot{\rho}$ whose value depends on the initial conditions assigned.

The remaining two equations, 4th and 5th row in (6.16) show that components of the constraint forces along the x_G and y_G directions are also affected by the moments of the inertia forces along the x_G and y_G axes. These, in turn, depend on the angular speed and angular acceleration of the body through some proportionality coefficients which in turn depend on some of the body's products of inertia. We see from these equations that non-zero reaction forces are generated by the rotation of the body even in case the eccentricity of the body's c.o.m. is zero, unless the inertia products J_{Oxz}^{L}, J_{Oyz}^{L} are both equal to zero, which means ultimately that the axis of rotation $z_G = z_L$ is one of the principal axes of inertia of the body.

We therefore conclude that the rigid body rotor is balanced, i.e. generates zero constraint forces while rotating, provided that the following two conditions are both satisfied:

1. the body's c.o.m. is located on the axis of rotation;
2. the axis of rotation is one of the principal axes of inertia for the body.

References.

[1] Shabana A., *Dynamics of Multibody Systems*, Ed. J. Wiley & Sons, New York, 1989.